# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – Short Answers

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| **Course Code:** | ITA03 | **Course Title:** | Mobile computing |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 2 – Short Answers |
| **Weightage:** | 30 % of CIA | **Total Marks:** | 100 Marks |
| **CO1 Statement:** | *Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.* | | |
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| **Section / Batch:** | **2023** | **Date:** | **13/10/2026** |

**Code of Conduct**

*I Dharani Nivash A/192324280 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dharani Nivash A

**Instructions**

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

| Faculty In-charge | Course Coordinator | Head of Department |
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| Signature: _____ | Signature: _____ | Signature: _____ |

| Date: _____ | Date: _____ | Date: _____ |
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## Questions

**Q1.** A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

**Q2.** A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

**Q3.** A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

**Q4.** During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

**Q5.** A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

**Q6.** An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7- cell pattern. How many channels does a single cell end up with?

**Q7.** A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

**Q8.** Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

**Q9.** A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

**Q10.** A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

**Q11.** A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

**Q12.** Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

**Q13.** A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

**Q14.** A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

**Q15.** A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

**Q16.** A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

**Q17.** Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

**Q18.** Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

**Q19.** A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing

distance, roughly how many handoffs would occur over an hour of travel?
**Q20.** A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

**Q21.** Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as $R^2$). Roughly how many cells will the rollout require?

**Q22.** An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

**Q23.** A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

**Q24.** A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

**Q25.** A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

**Q26.** A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

**Q27.** A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

**Q28.** A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

**Q29.** A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

**Q30.** An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

**Q31.** A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

**Q32.** Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

**Q33.** A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

**Q34.** A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

**Q35.** A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

**Q36.** A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

**Q37.** A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

**Q38.** A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

**Q39.** A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

**Q40.** A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?
**Q41.** A single base station covers 2.5 $km^2$ of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

**Q42.** An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

**Q43.** A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

**Q44.** A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

**Q45.** A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

**Q46.** A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

**Q47.** A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

**Q48.** A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

**Q49.** A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

**Q50.** A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = $\pi R^2$) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

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# CO1 – AT-2
# SHORT ANSWERS

2. Channels per cell: 560 / 7 = 80
(Channels per cell = Total channels / Cluster size)
**Total 80 Channels per Cell.**

Total GSM Voice Channels: 100 * 8 = 800
(Total channels = Number of carriers * Time slots per carrier)
**Total 800 Voice Channels.**

5. Carrier to Interference Ratio: -60 - (-92) = 32 dB
(C/I = Carrier power - Interference power)
**C/I = 32 dB.**

9. Total Channels in a Cluster: 4 * 15 = 60
(Total channels = Cells * Channels per cell)
**Total 60 Channels.**

11. TDMA Time Slot Duration: 5 / 8 = 0.625 ms
(Slot duration = Frame duration / Number of slots)
**Slot Duration is 0.625 ms.**

12. Number of Carrier Frequencies: 24000 / 200 = 120
(Number of carriers = Total bandwidth / Carrier spacing)
**Total 120 Carriers.**

15. Cluster Call-Handling Capacity: 6 * 250 = 1500
(Total capacity = Number of cells * Calls per cell)
**Total 1500 Simultaneous Calls.**

17. Total Bandwidth of a Base Station: 150 * 25 = 3750 kHz
(Total bandwidth = Number of channels * Bandwidth per channel)
**Total Bandwidth is 3.75 MHz.**

18. Total Guard-Band Bandwidth: (30 - 1) * 40 = 1160 kHz

(Total guard bandwidth = (Number of bands - 1) * Guard-band width)
**Total Guard-Band Bandwidth is 1.16 MHz.**

20. Total Network User Capacity: 8 * 40 * 250 = 80,000
(Total users = Regional centers * Base stations per center * Users per station)
**Total Capacity is 80,000 Users.**

22. Number of FDMA Channels: 6000 / 30 = 200
(Number of channels = Total bandwidth / Bandwidth per channel)
**Total 200 Channels.**

23. Offered Traffic in Erlangs: (1200 * 4 * 2.5) / 60 = 200
(Offered traffic = Subscribers * Calls per hour * Call duration in minutes / 60)
**Total Offered Traffic is 200**

24. Maximum Simultaneous Users: 40000 / 400 = 100
(Number of users = Total bandwidth / Bandwidth per user)
**Maximum 100 Simultaneous Users.**

25. Total Network User Capacity: 600 * 1200 = 720,000
(Total capacity = Number of base stations * Users per station)
**Total Capacity is 720,000 Users.**

26. Number of Paging Rounds: 200 / 4 = 50
(Paging rounds = Total cells / Cells searched per round)
**Total 50 Paging Rounds.**

28. Number of Roaming Subscribers: 600000 * 15 / 100 = 90,000
(Roaming subscribers = Total subscribers * Roaming percentage / 100)
**Total 90,000 Roaming Subscribers.**

29. Switching Center Utilization: (9000 / 12000) * 100 = 75%
(Utilization = Current load / Maximum capacity * 100)
**Total Utilization is 75%.**

31. Number of GSM Channels: 30000 / 200 = 150
(Number of channels = Total bandwidth / Channel spacing)
**Total 150 Channels.**

32. Successful Handoffs: 2500 * 99 / 100 = 2475
(Successful handoffs = Total attempts * Success percentage / 100)
**Total 2475 Successful Handoffs.**

34. Total User Capacity of a Cluster: 7 * 900 = 6300

(Total users = Number of cells * Users per cell)
**Total Capacity is 6300 Users.**

35. TDMA Slot Duration: 4.8 / 8 = 0.6 ms
(Slot duration = Frame duration / Number of slots)
**Slot Duration is 0.6 ms.**

38. Combined Cluster User Capacity: 9 * 600 = 5400
(Total users = Number of cells * Users per cell)
**Total Capacity is 5400 Users.**

39. Dropped Calls per Second: 60000 * 0.4 / 100 = 240
(Dropped calls = Total calls * Drop percentage / 100)
**Total 240 Dropped Calls per Second.**

41. Total Geographic Coverage: 2.5 * 150 = 375
(Total area = Area per base station * Number of base stations)
**Total Coverage is 375 km².**

43. Number of Paging Cycles: 150000 / 6000 = 25
(Paging cycles = Total subscribers / Users processed per cycle)
**Total 25 Paging Cycles.**